# The Axiom of Choice

Some objects are built one piece at a time. A union of countably many countable sets is listed by listing each set, and listing each set means picking a list for each of infinitely many sets, with no rule to say which. An injection, a set of representatives and a total order are built the same way: one pair, one class, one comparison at a time. When the pieces can be counted off, induction finishes the job, one step for each natural number. When they cannot, the words "and so on" stand in for a step that is never given, and that is not a proof. *Is it enough to say "choose one from each"?*

The chapter answers in three moves. Section 6.1 states the principle that allows infinitely many arbitrary choices at once, the **axiom of choice**, shows what is and is not a choice (a choice made by a rule needs no axiom), and uses it to give every surjection a right inverse, every countable union of countable sets a listing, and every infinite set a copy of $\mathbb{N}$. Section 6.2 states Zorn's lemma (a non-empty partially ordered set in which every non-empty chain has an upper bound has a maximal element) and turns it into a single method, the maximality template: if every growing chain of partial objects has a partial object above it and every unfinished one can be enlarged, then a complete one exists. It runs the template to extend every partial order to a total one. Section 6.3 proves that Zorn's lemma gives the axiom of choice, settles the question Section 4.4 left open (any two sets are comparable in size), and proves the well-ordering theorem. The lemma itself is not proved here: it is equivalent to the axiom, and this volume proves only the half that gives choice.

**What you need.** *Logic and Proof*, Chapters 1–8, in particular induction (Logic and Proof, Axiom 4.2.1), the rule $\exists$E (Logic and Proof, Definition 5.6.1), the negation of bounded quantifiers (Logic and Proof, Proposition 6.2.3) and of definitions (Logic and Proof, Example (Three negations from analysis)), the irrationality of $\sqrt{2}$ (Logic and Proof, Theorem 2.2.1), and the catalogue of what is not allowed (Logic and Proof, Example (What is not allowed)), above all (N13), "an 'and so on' needs its step". From Chapters 1 and 2: parity and divisibility (Theorem 1.1.2, Definition 1.1.2), sets cut out by a property, power sets and families (Definition 1.2.3, Definition 1.3.2, Definition 1.3.3, Definition 1.3.4), relations (Definition 1.4.3), functions and their equality (Definition 2.1.1, Proposition 2.1.1, Proposition 2.1.2), surjections and their composites (Definition 2.2.2, Theorem 2.2.1), the correspondence between subsets and 0–1 functions (Theorem 2.3.5), left and right inverses (Definition 2.4.1, Theorem 2.4.1, Proposition 2.4.2), and preimages (Definition 2.5.2). From Chapter 3: the arithmetic of $\mathbb{N}$ (Lemma 3.1.1), well-ordering (Theorem 3.1.3) and induction (Theorem 3.2.1), definition by recursion (Theorem 3.3.1), the absence of an infinite descent (Example (No endless descent in the natural numbers)), the largest element of a finite set (Proposition 3.3.8), and the method of choosing the form of a proof (Example (Method: choosing direct, contrapositive or contradiction)). From Chapter 4: finite sets and the pigeonhole principle (Definition 4.1.2, Corollary 4.1.7, Theorem 4.1.9, Corollary 4.1.3, Proposition 4.1.6, Example (Hilbert's Hotel), Proposition 4.1.5, Proposition 4.1.1), countable sets (Definition 4.2.1, Definition 4.2.2, Theorem 4.2.2, Theorem 4.2.6, Corollary 4.2.7, Proposition 4.2.4, Proposition 4.2.3, Proposition 4.2.5, Theorem 4.2.8), the uncountability of $\mathbb{R}$ (Theorem 4.3.4) and $\mathbb{R}$ equinumerous with an open interval (Example (The real line and its intervals)), and comparison of sizes (Definition 4.4.1, Theorem 4.4.3, Proposition 4.4.2, Proposition 4.5.1). From Chapter 5: equivalence classes and partitions (Definition 5.2.2, Theorem 5.2.1, Example (The reals modulo the integers)) and orders (Definition 5.3.1, Proposition 5.1.1, Example (Inclusion is a partial order, rarely total), Definition 5.3.2, Definition 5.3.4, Proposition 5.3.2, Example (Maximal elements that are not greatest), Proposition 5.3.1, Definition 5.3.6).

**Time.** About 20 hours on the core track (all of A, one B and one C per section, the exit checks); about 30 hours for everything.

## 6.-1.1 Roadmap

- **The axiom of choice.** A union of countably many lists and the choice it hides; the axiom of choice and its statement; when no axiom is needed, because a rule picks the element (a finite number of choices, a least element, a formula); a right inverse for every surjection; surjections as a way of comparing sizes; countable unions of countable sets; every infinite set contains a copy of $\mathbb{N}$, so $\mathbb{N}$ is the smallest infinite size; and an infinite set is equinumerous with a proper part of itself.
- **Zorn's lemma and how to use it.** A pair-by-pair matching that cannot finish; chains in a partially ordered set, and why finitely many elements of the union of a chain of sets lie in one member; Zorn's lemma, why each hypothesis is needed, and why it is stated and not proved; the maximality template, which gives a set of representatives for any equivalence relation; and extending a partial order to a total order, one added comparison at a time.
- **Equivalent forms of choice.** Zorn's lemma gives the axiom of choice; any two sets are comparable, so sizes satisfy trichotomy; every set can be well-ordered, and a well-ordered set has a choice function; and a table of which implications this volume proves and which *Axiomatic Set Theory* proves.

## 6.-1.2 Named moves

- **Name the choice** (Example (Method: name the choice)). Before using the axiom, look for the rule: one choice needs no axiom, finitely many are made one at a time, and a least element or a formula picks without help. If there is no rule, say that the axiom is used and where. (Section 6.1)
- **The maximality template** (Example (Method: the maximality template)). Partial objects ordered by extension; the union of a chain is again a partial object, since each defining condition mentions finitely many elements and these lie in one member of the chain (Lemma 6.2.1); Zorn's lemma gives a maximal one; a maximal one is complete, since a missing piece would give a strictly larger partial object. (Section 6.2, used again in Section 6.3)

## 6.-1.3 Threads

- **Right inverses.** Section 2.4 proved that a right inverse forces surjectivity and could not build one. Section 6.1 does (Theorem 6.1.2), and shows that this statement is equivalent to the axiom of choice.
- **Choice.** Sections 4.2, 4.4 and 4.5 avoided the axiom by giving a rule each time: least preimages, a surjection built from an injection, a copy of $\mathbb{N}$ supplied in advance. Section 6.1 settles the three cases where no rule exists. Section 6.3 derives the axiom from Zorn's lemma (Theorem 6.3.1) and from well-ordering (Proposition 6.3.5), and *Axiomatic Set Theory*, Chapter 4 proves the converse.
- **Comparing sizes.** Section 4.4 asked whether any two sets are comparable. Section 6.2 opens with a pair-by-pair matching that does not finish, and Section 6.3 finishes it with Zorn's lemma (Theorem 6.3.2).
- **Well-ordering.** Section 3.1 proved that every non-empty set of natural numbers has a least element, and Section 5.3 named the sets with that property. Section 6.3 proves that every set can be given such an order (Theorem 6.3.4).

## 6.-1.4 Summary

The axiom of choice (Axiom 6.1.1) gives a function that picks one element from each of a family of non-empty sets. It is equivalent to the statement that every surjection has a right inverse (Theorem 6.1.2); it makes a countable union of countable sets countable (Theorem 6.1.4), so that $|A| \leq |B|$ exactly when there is a surjection $B \to A$ (Proposition 6.1.3); and it gives every infinite set a countably infinite subset (Theorem 6.1.5), so that $\mathbb{N}$ is the smallest infinite size and an infinite set is equinumerous with a proper subset (Theorem 6.1.6). A choice made by a rule needs no axiom (Example (Method: name the choice)).

A chain is a subset of a partially ordered set on which the order is total (Definition 6.2.1), and finitely many elements of the union of a chain of sets lie in one member (Lemma 6.2.1). Zorn's lemma (Theorem 6.2.2): a non-empty partially ordered set in which every non-empty chain has an upper bound has a maximal element, usually one of many and not a greatest one. It is stated without proof; it implies the axiom of choice (Theorem 6.3.1), and the converse belongs to *Axiomatic Set Theory*, Chapter 4. The maximality template (Example (Method: the maximality template)) gives a set of representatives for any equivalence relation (Example (A set of representatives)), extends every partial order to a total order (Theorem 6.2.4), by adding one comparison at a time (Lemma 6.2.3), shows that any two sets are comparable in size (Theorem 6.3.2), so that exactly one of $|A| < |B|$, $|A| = |B|$, $|B| < |A|$ holds (Corollary 6.3.3), and well-orders every set (Theorem 6.3.4), which in turn gives choice (Proposition 6.3.5).

Every object built in this chapter is shown to exist and none is written down: no formula is known for a set of representatives of $\mathbb{R}$ modulo $\mathbb{Q}$, for an injection between two arbitrary sets, for a well-order of $\mathbb{R}$, or for a total order containing $\subseteq$ on $\mathcal{P}(\mathbb{R})$. Where a formula exists, as for divisibility on $\mathbb{N}$ (Example (A total order containing divisibility)), it is the better proof.

## 6.-1.5 Terms

- **axiom of choice**: for every family of non-empty sets indexed by a set $I$, there is a function that picks an element of each (Axiom 6.1.1).
- **chain**: a subset in which any two elements are comparable (Definition 6.2.1).
- **choice function**: a function $c$ on the index set with $c(i) \in A_i$ for every $i$ (Axiom 6.1.1).
- **linear extension**: a total order containing a given partial order; "linear order" is a synonym of "total order" (Theorem 6.2.4).
- **maximality template**: partial objects, unions of chains, Zorn's lemma, completeness of a maximal object (Example (Method: the maximality template)).
- **partial choice function**: a set of pairs $(i, x)$ with $x \in A_i$, at most one for each index (Theorem 6.3.1).
- **partial matching**: a relation $G \subseteq A \times B$ in which each element of $A$ and of $B$ occurs in at most one pair (Theorem 6.3.2).
- **set of representatives**: a set meeting every equivalence class in exactly one element (Example (A set of representatives)).
- **well-order**: a total order in which every non-empty subset has a least element (Definition 5.3.6).
- **Zorn's lemma**: a non-empty partially ordered set in which every non-empty chain has an upper bound has a maximal element (Theorem 6.2.2).